

Self-Stratifying Silicone Coating with Nonleaching Antifoulant for Marine Anti-Biofouling

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Chemical immobilization of antifoulant groups to polymers is a promising approach to develop ecofriendly antifouling materials with static antifouling ability and a durable efficacy. In this work, telomer of dodecafluoroheptyl methacrylate (DFMA), triclosan acrylate (TA), and 3-mercaptopropyl trimethoxysilane (KH590) is grafted to bis-silanol terminated poly(dimethylsiloxane) (PDMS), yielding a novel self-stratifying silicone coating containing non-leaching antifoulants. X-ray photoelectron spectroscopy (XPS) analysis reveals that the antifoulant groups are enriched on the coating surface due to DFMA migration. Such a modified PDMS coating can significantly inhibit the growth of marine bacterial biofilm and the adhesion of diatoms owing to the nonleaching antifoulant groups. On the other hand, the coating exhibits a good fouling release performance since it inherits the low surface energy ($\approx 20 \text{ mJ m}^{-2}$) and low elastic modulus ($< 2 \text{ MPa}$) from PDMS forming a physically dynamic surface. Marine field tests demonstrate that the coating has excellent antifouling ability after immersion in natural seawater for more than 6 months.

dynamic surfaces,^[4] polymethacrylate bearing ferrocenyl groups,^[5] amphiphilic copolymers,^[6] liquid-infused slippery surfaces,^[7] surface-deforming materials,^[8] enzyme containing coatings,^[9] and so on. Among them, silicone-based FRCs with excellent fouling release ability have gained considerable attention because it is environment-friendly and has drag reducing ability.^[10]

The fouling release ability of silicone elastomers arises from its low surface energy and low elastic modulus.^[11] The latter makes the elastomer form a physically dynamic surface to which microorganisms weakly adhere.^[2b,12] Yet, an external shearing force is needed to detach fouling organisms from the surface of silicone elastomer that cannot resist biofouling in static immersion without strong water flow. Moreover, it was reported that slime consisting of bacteria and diatoms

is able to accumulate on silicone surface, and it is difficult to remove even with the help of water jet.^[13] Therefore, the fouling resistance of silicone-based FRC needs improving, particularly that on static conditions.

Physical addition of antifoulant can improve the antifouling ability of silicone, but the release of antifoulant is usually not consistent since silicone is not erodible in seawater. Moreover, it may affect the metabolism and growth of nontarget marine organisms. Chemically immobilization of biocidal moieties such as organic antifoulant,^[14] quaternary ammonium salt,^[15] zwitterion,^[16] and amphiphile^[17] on PDMS elastomer can avoid their release. Yet, most of such biocidal moieties are readily trapped in bulk and cannot fully exert fouling resistance. Moreover, their presence often increases the modulus and swelling of the coating, leading to decreased fouling release and mechanical performances. Enrichment of the grafted biocidal moieties on coating surface can lead to good antifouling performance without undermining the advantages of silicone elastomers. However, it is a challenge because silicone segment tends to cover biocidal moieties during the formation of coating due to its low surface energy.

In this study, we have prepared a novel telomer of triclosan acrylate (TA), dodecafluoroheptyl methacrylate (DFMA), and 3-mercaptopropyl trimethoxysilane, where triclosan is a broad spectrum antimicrobial agent.^[18] Silicone-based coating is

1. Introduction

Marine biofouling arises from the accumulation of marine microorganisms, plants, and animals on immersed surfaces.^[1] It causes many negative consequences on marine industry and maritime activities, such as increased frictional resistance and fuel consumption of ships, accelerated corrosion of metal and blocked seawater pipelines in near-sea energy infrastructures.^[2] The global ban of effective but highly toxic tributyl-tin (TBT) based self-polishing coatings urges the development of ecofriendly antifouling systems, including silicone-based fouling release coatings (FRC),^[3] degradable polymers with

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prepared by crosslinking the telomer with bis-silanol terminated poly(dimethylsiloxane) (PDMS) via condensation reaction at room temperature. The fluorinated moiety incompatible with PDMS has strong tendency to migrate onto surface during the formation of the coating, yielding to a self-stratifying system with antifoulant groups enriched on the surface. Thus, the coating is expected to exhibit both fouling resistance and fouling release, i.e., to inhibit fouling adhesion and facilitate its removal. Even more, the chemical immobilization of antifoulants makes the coating durable and ecofriendly. We have investigated the antifouling performance on antibacterial biofilm, antidiatom, and pseudobarnacle. The long-term antifouling efficacy was evaluated by marine field test. We attempt to develop ecofriendly antifouling system with durable efficiency in static conditions.

2. Results and Discussion

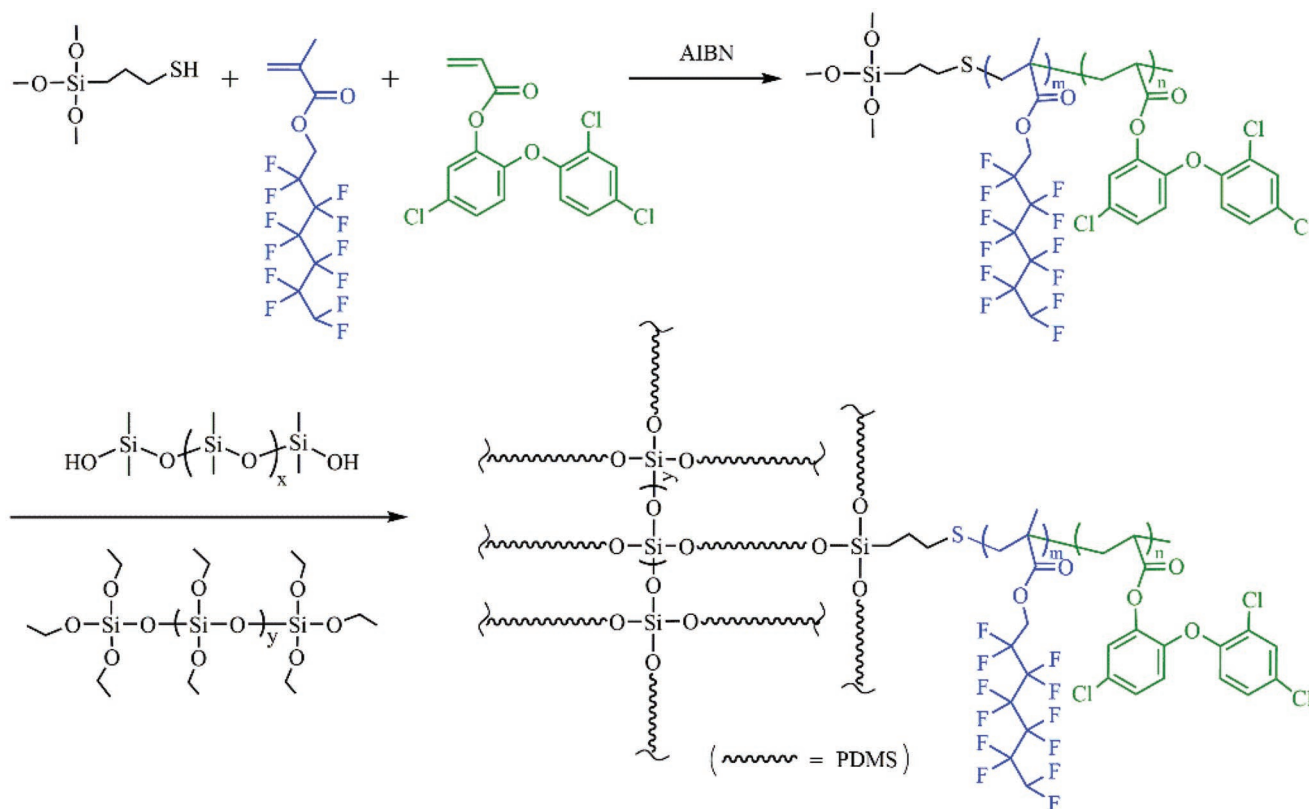
2.1. Preparation of the Coatings

Scheme 1 shows the synthesis of DFMA-TA-KH590 telomer and its grafting to the PDMS network. The successful synthesis of the telomer was determined by ^1H NMR (Figure S1, Supporting Information), where the molar ratio is 5.1/4.3/1 for DFMA/TA/KH590. More characterization data can be found in Table S1 (Supporting Information). The telomer is designated as PFTA for convenience. PFTA was subsequently grafted to a room-temperature-vulcanized (RTV) PDMS elastomer,

generating modified silicone coating (PDMS-FTx). Another PDMS-based coating (PDMS-Tx) containing telomer of TA and KH590 (molar ratio: 9.3/1, designated as PTA) was also prepared to extract the role of DFMA on the surface and bulk properties of the resulting coatings **Table 1**.

2.2. Surface Composition

Figure 1 shows the XPS spectra of PDMS-based coatings. The peaks of Si (2s and 2p) at 150.5 and 100.4 eV in the scan spectra of all coatings arise from the PDMS segments. The peak at 685.7 eV attributed to F (1s) in spectra of PDMS-FTx indicates the presence of dodecafluoroheptyl groups on the surface. The peak for Cl cannot be observed on PDMS-T4 even on the high resolution spectra, indicating that no TA is located on the surface even if the coating contains 4 wt% of TA in bulk. In contrast, such a peak appears in the spectra of PDMS-FTx containing 2–4 wt % of TA. Clearly, TA is dragged upward by fluoro groups, leading to a self-stratifying coating. **Table 2** shows the data of elemental weight ratio. For PDMS-FTx, F/Si and Cl/Si values on the surface are higher than those in bulk, namely, the surface is enriched by PFTA. We also conducted the energy dispersive X-ray spectrometer (EDS) analysis on the cross-section of PDMS-based coatings (Figure S2, Supporting Information). For PDMS-T4, the Cl element is evenly distributed in the cross-section, indicating that most of the antifoulant is buried in bulk. In comparison, the Cl element of PDMS-FT12 shows a gradient distribution with higher content on the surface, which



Scheme 1. Synthesis of PFTA and PDMS-FTx.

Table 1. Composition of PDMS-based coatings.

Sample	Bis-silanol PDMS [g]	PFTA/PTA ^{a)} [g]	ES-40 [g]	DFMA content [%]	TA content [%]
PDMS	10.00	00	0.25	0	0
PDMS-T4	10.00	0.40	0.25	0	3.7
PDMS-FT4	10.00	0.40	0.25	2.2	1.6
PDMS-FT8	10.00	0.40	0.25	4.2	3.0
PDMS-FT12	10.00	0.40	0.25	6.1	4.4

^{a)} PFTA was added in PDMS-T4 and PFTA was added in PDMS-FTx.

demonstrates that the antifoulant is self-enriched on the surface, further confirming that the coating is a self-stratifying system.

2.3. Surface Wettability and Elastic Modulus

We studied the surface wettability of PDMS-based coatings (Figure 2a,b). PDMS shows low surface energy of around 22 mJ m^{-2} , a characteristic property of a silicone network. PDMS-T4 has a similar surface energy to the one of PDMS because the antifoulant groups cannot migrate to surface to make notable difference in wettability. PDMS-FTx coatings have a surface energy slightly lower than those of PDMS and PDMS-T4, indicating that the surface-enriched dodecafluoroheptyl group decreases the surface energy of the coating. For each coating, the surface is hydrophobic with WCA above 105° , and the hydrophobicity slightly varies after immersion in artificial seawater (ASW). These facts indicate that composition of the surface is stable. This is beneficial for fouling release property of the coatings.

Figure 2c,d shows the bulk and surface modulus of PDMS-based coatings from DMA and AFM tests (storage modulus-strain and force-depth curves shown in Figure S4 of the Supporting Information). For PDMS, the surface modulus is close to the bulk modulus. It has the lowest bulk and surface modulus among the samples. Samples containing the grafted telomer have higher bulk and surface elastic modulus than

PDMS. The T_g values of the telomers PFTA and PTA reported in Table S1 of the Supporting Information are higher than a PDMS elastomer which is close to -100°C . These data demonstrate that these polymer chains exhibit less mobility in comparison with the PDMS chains due to the steric hindrance of the phenyl groups of TA monomer units. Note that PDMS-FT4 has a surface modulus higher than the unmodified PDMS though their bulk modulus is very close. This is because PFTA distributed on the surface

makes the upper layer rigid whereas such PFTA content is not enough to change the bulk property. Anyhow, the modulus is lower than 1 MPa for each sample, indicating that the grafted telomer has a slight effect on low elastic modulus PDMS network within the range of its content in this study. This is in favor of favoring the detachment of fouling organisms from the coating surface.

2.4. Antifouling and Fouling Release Performances

2.4.1. Leaching of Toxic Substance

We firstly examined whether the PDMS-based coatings release grafted antifoulant or other toxic substances. For comparison, we also tested the PDMS coating with 4 wt% of physically added antifoulant. Diatom, a typical model microorganism, was selected because its growth is significantly inhibited by small amount of toxic substance.^[3,19,20] Figure 3 shows the chlorophyll content of diatoms after incubation using an F/2 medium containing the leachate from the coatings after immersion for 14 d. The chlorophyll content from medium with PDMS leachate is $0.7 \mu\text{g mL}^{-1}$, close to the one from medium without leachate, indicating that the PDMS coating does not release any toxic substances toward the diatom *Navicula incerta*. In comparison, the chlorophyll content of the PDMS coating containing the physically added antifoulant is lower because the antifoulant is released during the immersion and inhibits the growth of diatoms. The chlorophyll content from medium with PDMS-FTx leachate is similar to that of PDMS, which shows that the grafted antifoulant is a nonleaching antifoulant.

To quantify the leached substance at a longer period, we performed mass loss test (Figure S2, Supporting Information). Pure PDMS has a small mass loss at the early stage due to the leaching of residue solvent, unreacted silicone oil, and catalyst. As immersion prolongs up to 3 months, its mass almost does

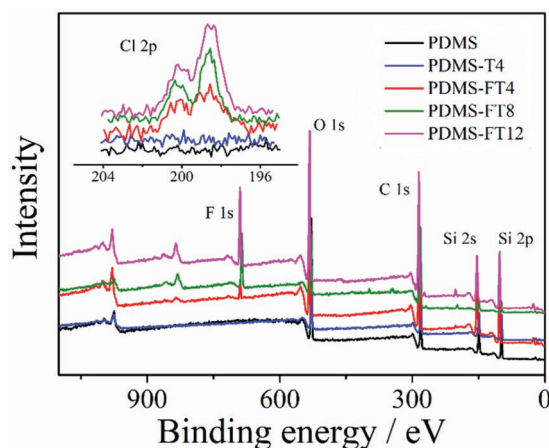


Figure 1. XPS scan spectra of PDMS-based coatings. The inset shows the Cl spectra at high resolution.

Table 2. Elemental weight ratio of Si/F/Cl on PDMS-based coatings.

Sample	Si/F/Cl in bulk	Si/F/Cl on surface
PDMS	100/0/0	100/0/0
PDMS-T4	100/0/3.2	100/0/0
PDMS-FT4	100/3.4/1.3	100/13.7/2.1
PDMS-FT8	100/6.8/2.5	100/36.4/7.5
PDMS-FT12	100/10.1/3.9	100/75.7/12.4

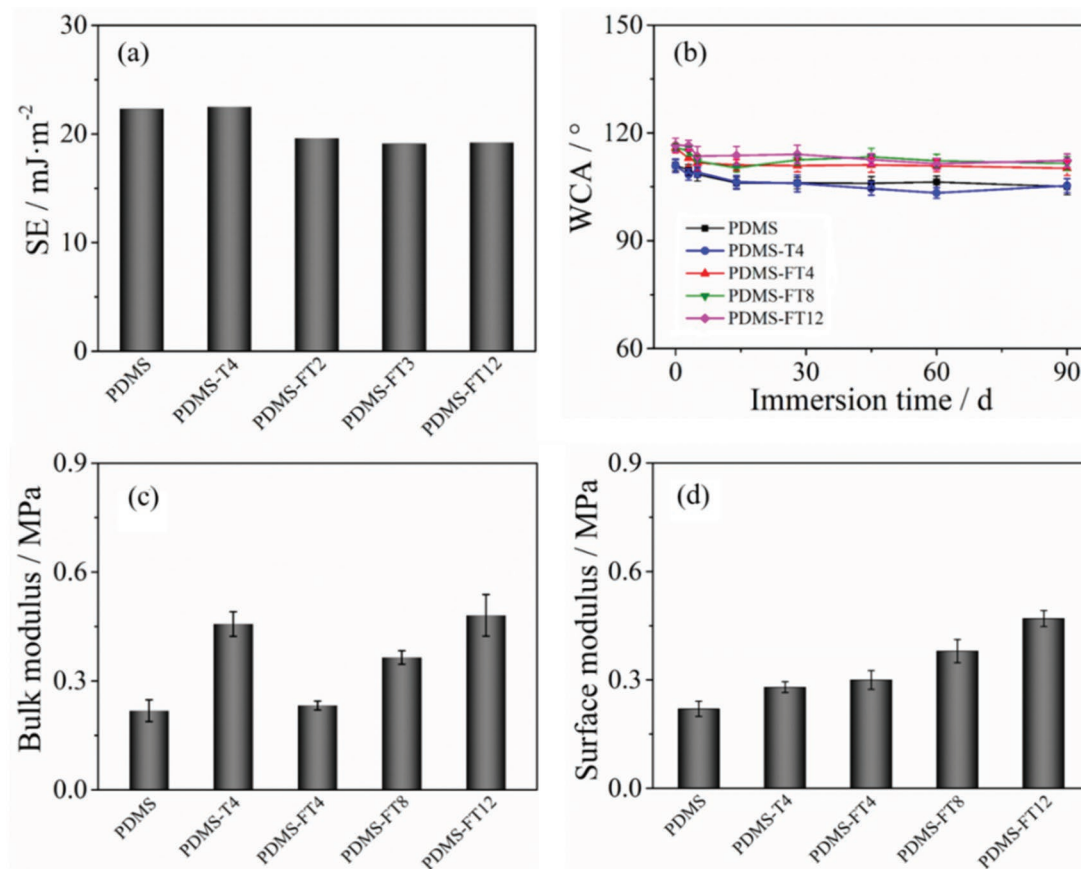


Figure 2. a) Surface energy (SE), b) time dependence of water contact angle (WCA) during immersion in ASW, c) bulk modulus, and d) surface modulus of PDMS-based coatings.

not change. PDMS-FT12 shows a similar mass loss behavior to that of PDMS, indicating PFTA is not released to seawater (or the release is extra low). In contrast, PDMS with physically

mixed triclosan exhibits a quick and large mass loss, which is mainly due to the release of unbound triclosan. Actually, Chisholm and co-workers^[19] also reported copolymer containing triclosan acrylate did not release triclosan in seawater. This is probably because the large steric hindrance of triclosan leads to slow hydrolysis of ester group. The mass loss together with the chlorophyll tests prove triclosan is nonleaching. In brief, the system should not pose threat to the marine environment.

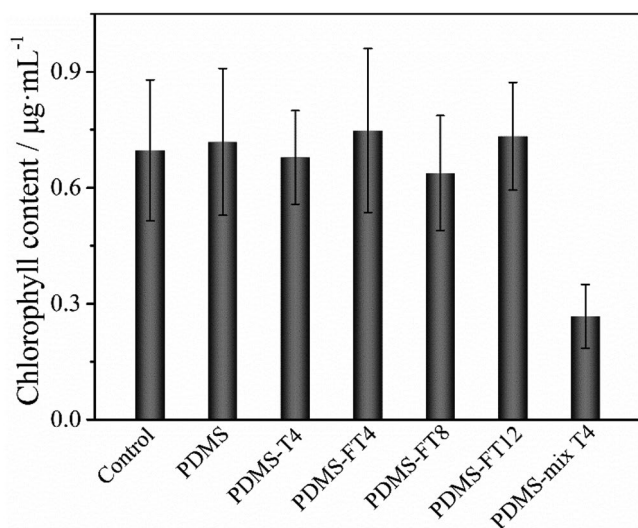


Figure 3. The chlorophyll content of diatom cultured with F/2 medium containing leached substance from PDMS-based coating (PDMS-mix T4 refers to PDMS with 4 wt% of physically added triclosan) or fresh medium as control.

2.4.2. Anti-biofilm Test

Then, we studied the performance of coatings to inhibit marine bacterial biofilm growth of *Pseudomonas* sp. (Figure 4a). The biofilm is stained with crystal violet and the absorbance is proportional to the amount of biofilm on the coating.^[20] On a polystyrene (PS) control surface, the bacterial biofilm accumulates with an absorbance of 2.1, whereas the biofilm on unmodified PDMS surface has an absorbance of 2.2, suggesting that PDMS is not able to resist biofilm formation. PDMS-T4 has a poor fouling resistance close to the one of PDMS because the antifoulant telomer is buried in bulk (Figure 1). In contrast, much less biofilm grows on PDMS-FT4 surface. The sample has a lower antifoulant content in bulk than that of PDMS-T4 but exhibits higher antibacterial efficiency, indicating that the surface enrichment of antifoulant plays a critical role in

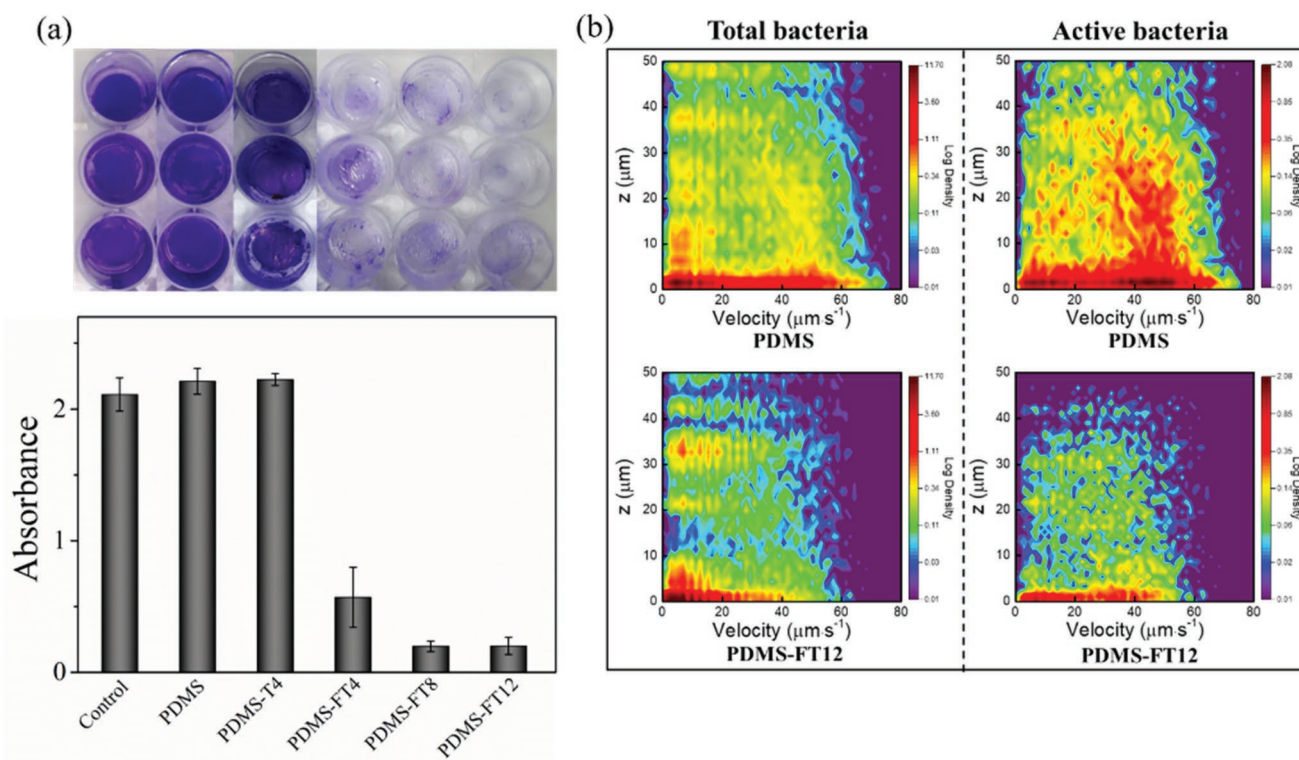


Figure 4. a) *Pseudomonas* sp. biofilm grown on PDMS-based coatings stained with crystal violet and the absorbance of crystal violet from optical density test; b) Density distribution of instantaneous 3D velocities and distances from the surface (z) for *Pseudomonas* sp. upon PDMS and PDMS-FT12. Color bars represent the relative density of data points with a logarithmic scale.

inhibiting the biofilm formation. As the antifoulant content on coating surface increases, the antibacterial ability increases. As discussed above, the antifoulant is nonleaching, so the bacteria are not driven away by the leached triclosan but grafted one.^[13,19]

To insight into the mechanism of action of the PDMS-FTx, we tracked 3D real-time motion behavior of *Pseudomonas* sp. near the coating surface (Figure 4b). Clearly, the amount of total bacteria moving near PDMS-FT12 surface is lower than that of PDMS. Furthermore, much less bacteria moves quickly near PDMS-FT12 surface than PDMS. The facts indicate that PDMS-FTx with nonleaching antifoulant behave as contact-active coatings. That is, when the marine bacteria are in contact with the antifoulant-enriched surface, they tend to escape from the surface.

2.4.3. Antidiatom Test

Figure 5 shows the adhesion density of diatom *Navicula incerta* on PDMS-based coatings. Unmodified PDMS has a limited antiadhesion ability with a diatom density of ≈ 350 cell mm^{-2} , slightly higher than the one found on the epoxy control surface. This is because diatoms tend to settle on hydrophobic surfaces,^[21] PDMS-T4 exhibits an antiadhesion efficiency similar to PDMS since the antifoulant groups are trapped in bulk. PDMS-FT4 has a considerably lower diatom density than PDMS and PDMS-T4, indicating that the antifoulant on the surface can effectively resist diatoms. PDMS-FT8 and PDMS-FT12 with high antifoulant contents on their surface

demonstrate excellent antiadhesion performances with a low diatom density of ≈ 50 cell mm^{-2} . It is well known that silicone elastomer coatings suffer from slime settlement consisting of bacteria and diatoms, especially in static conditions. Clearly, the grafted antifoulant significantly improves the resistance of the coating toward the growth of bacterial biofilm and the adhesion of diatoms.

2.4.4. Fouling Release Test

Figure 6 shows the removal strength of adhered pseudobarnacles on PDMS-based coatings. A high removal strength of 1.1 MPa is needed to detach pseudobarnacles from the epoxy control surface. In contrast, unmodified PDMS shows a remarkable fouling release property with a low removal strength of 0.1 MPa, close to reported data.^[22] Either PDMS-FT4 or PDMS-FT8 has a removal strength close to PDMS. It is well established that the combination of low surface energy and low elastic modulus is responsible for the fouling release ability. PDMS-FT4 and PDMS-FT8 have lower surface energy but slightly higher modulus than those of PDMS, so their fouling release performance is similar. PDMS-FT12 shows a higher removal strength of ≈ 0.2 MPa because it has a notably higher modulus than PDMS in spite of its lower surface energy. In this case, modulus has more influence on the fouling release property. Nevertheless, each sample has excellent and fouling release ability considering the low removal strength values.

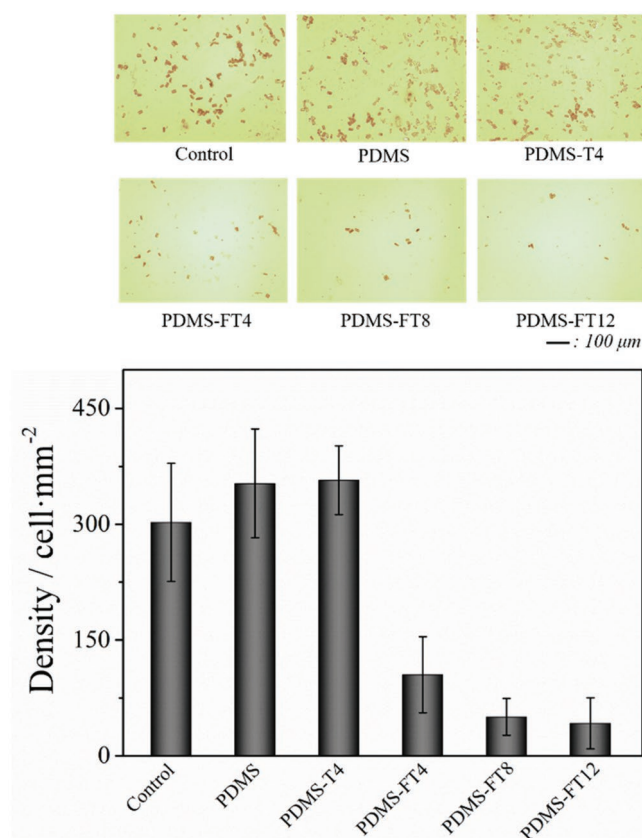


Figure 5. Optical images (200×) of diatom *N. incerta* adhered on PDMS-based coatings and the quantified diatom density.

2.4.5. Marine Field Test

We further evaluated the antifouling performance of PDMS-based coatings by marine field test (Figure 7). The coatings has reasonable adhesion to the epoxy substrate similar to that of commercial PDMS coatings which should meet the

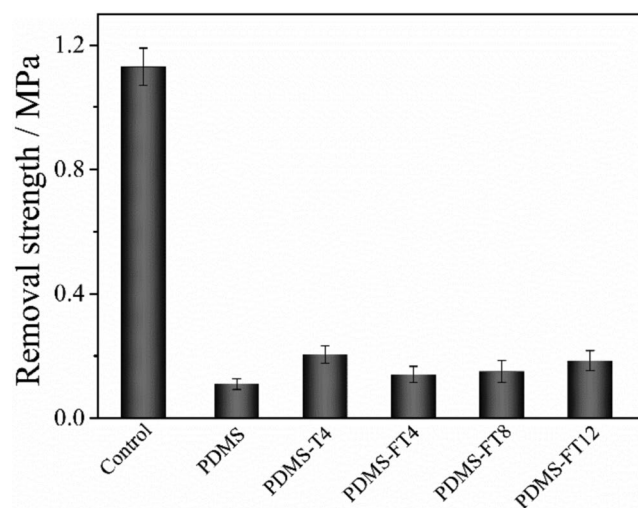


Figure 6. Removal strength of pseudobarnacles adhered on PDMS-based coatings.

requirement of practical applications. After immersion in natural seawater for 3 months, the epoxy panel control was completely covered by marine organisms mainly consisting of bryozoan and algae. Unmodified PDMS does not exhibit good resistance to organisms due to the lack of antifouling moieties. PDMS-T4 also has a poor antifouling ability because most of the antifoulant is trapped in the bulk of the coatings. However, PDMS-FT4, PDMS-FT8, and PDMS-FT12 exhibit high efficiency, where only few organisms are attached. These results show that the surface-enriched antifoulants play an important role also on macroorganisms. Applying a water jet to the coating surface can wash most adhered organisms away, indicating that the coating has good fouling release ability due to the presence of soft PDMS segments. After 6 months, abundant organisms such as barnacles, ascidian, and algae accumulate on the epoxy surface as the fouling pressure becomes even more severe. PDMS and PDMS-T4 are seriously fouled as well, but less organisms on PDMS-FTx are observed. Moreover, the PDMS-FT8 and PDMS-FT12 surfaces are almost clean after water jet treatment. Clearly, the combination of fouling resistance and fouling release properties leads to the long-term antifouling performance of PDMS coatings containing grafted telomer.

3. Conclusion

We have prepared a novel self-stratifying silicone coating with nonleaching antifoulant by grafting telomer of DFMA, TA, and 3-mercaptopropyl trimethoxysilane (KH590) into bis-silanol terminated PDMS. The coating shows good fouling release performance and resistance to marine bacterial biofilm and diatoms. Marine field tests demonstrate the excellent static and durable antifouling ability in seawater of TA-enriched PDMS coating surface. The surface-enriched antifoulant groups are responsible for the antifouling performance, whereas the low surface energy and low modulus yielding physically dynamic surface provide fouling release performance to the coatings. The chemical attachment of TA to the PDMS network prevents its release into the marine environment. The coatings are expected to be used in marine anti-biofouling applications.

4. Experimental Section

Materials: PDMS and poly(diethoxy siloxane) (ES-40) were purchased from Gelest and used as received. DFMA was purchased from Dayang Chemical and used as received. Dibutyltin diacetate (DBTDA), 3-mercaptopropyl trimethoxysilane (KH590), and ethyl acetate (EA) were purchased from Aladdin and used as received. Azobisisobutyronitrile (AIBN) from Aladdin was purified by recrystallization from methanol. Tetrahydrofuran (THF) from Sinopharm was refluxed over CaH₂ and distilled prior to use. TA was synthesized following a procedure reported before.^[9] Artificial seawater was prepared following ASTM D1141-98 (2003).

Synthesis of Fluorocarbon Acrylate-Antifoulant Telomer: The telomerization of DFMA, TA, and KH590 was performed at the molar feeding of 4:4:1 (Scheme 1). Typically, 4.00 g of DFMA (10.0 mmol), 3.44 g of TA (10.0 mmol), 0.49 g of KH590 (2.5 mmol), and 0.02 g of AIBN (0.25 wt % of the reactants) were dissolved in 20.0 mL of THF and added to a 100 mL Schlenk tube with a magnetic stirring bar.

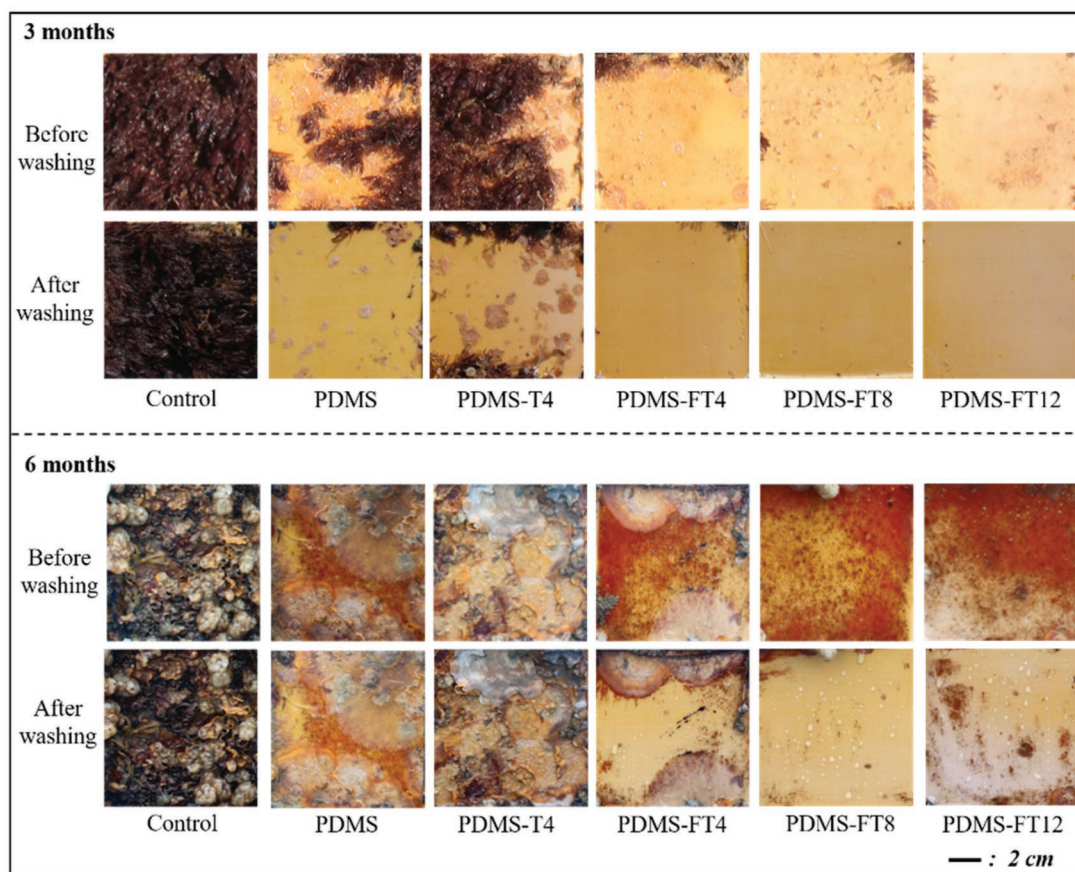


Figure 7. Pictures of PDMS-based coatings after immersion in Xiamen Bay, China before and after a water jet treatment (pressure: 2 MPa).

The solution was degassed by three freeze-evacuate-thaw cycles and then heated at 70 °C under argon for 24 h. The yielded solution was precipitated in hexane, dried under vacuum at 40 °C overnight, and stored under argon. For convenience, this telomer is designated as PFTA. Another telomer without DFMA was synthesized at a molar ratio of 8:1 (TA: KH590) by the same procedure and designated as PTA. The ^1H -NMR spectra of PFTA and PTA with assigned protons are shown in Figure S1 (Supporting Information). The characterization data are summarized in Table S1 (Supporting Information).

Preparation of Crosslinked PDMS Coating Containing PFTA: PFTA was grafted to PDMS by a condensation reaction (Scheme 1 and Table 2). Typically, 10.00 g of bis-silanol terminated PDMS, 0.80 g of PFTA, 0.25 g of crosslinker ES40, and 2.5 mg of catalyst DBTDA were dissolved in 30.0 mL of EA. The solution was stirred with a magnetic bar at room temperature for 30 min. Then the solution was applied on glass fiber reinforced epoxy panel, 24-well polystyrene (PS) microplate (only for bacterial biofilm test) and Teflon dish (only for dynamic mechanical analysis). Then, they are placed under a fume hood at room temperature for 5 d. The dry thickness of PDMS-based coatings are $\approx 200\ \mu\text{m}$ for all characterizations except dynamic mechanical analysis ($\approx 500\ \mu\text{m}$). For convenience, the coating containing PFTA is designated as PDMS-FTx, where x is the weight percentage of PFTA. Another crosslinked PDMS coating containing 4 wt% of PTA (designated as PDMS-T4) was prepared by the same procedure.

Characterization: Proton nuclear magnetic resonance spectroscopy (^1H -NMR) – All ^1H -NMR spectra were recorded on a Bruker AV600 NMR spectrometer using CDCl_3 as solvent and tetramethylsilane (TMS) as internal standard.

Gel Permeation Chromatography (GPC) – The number-average molecular weight (M_n) and the polydispersity (D_M) were determined

by Agilent gel permeation chromatography (1260 Infinity) with RI and UV detectors. The tests were conducted in THF at 35 °C using two identical PLgel columns (5 μm , MIXED-C) at a flow rate of 1.0 mL min^{-1} . Calibration was done with a series of narrowly dispersed polystyrene standards (1.30×10^3 – $2.21 \times 10^6\ \text{g mol}^{-1}$).

Differential scanning calorimetry (DSC) – The glass transition temperatures (T_g) of PFTA and PTA were measured with a Q100 differential scanning calorimeter (TA Instruments). The sample was heated to 150 °C to erase thermal history. Then it was cooled down to $-60\ ^\circ\text{C}$ and scanned from -60 to 120 °C at a heating rate of 10 °C min^{-1} . The T_g values were determined as the midpoint between the onset and the end of a step transition by TA Instruments Universal Analysis 2000 software.

X-ray photoelectron spectroscopy (XPS) – XPS analysis was conducted on an Axis Ultra DLD (Kratos Analytical) with a monochromate Al K α ($h\nu = 1486.6\ \text{eV}$) source at take-off angles of 90° under a pressure of 5×10^{-9} torr. The X-ray gun was operated at the voltage of 15 kV and current of 5 mA. Binding energies were all referenced to the C 1s at 284.6 eV.

Energy dispersive X-ray spectrometer (EDS) – The distribution of antifoulant in the cross-section of coating was measured in terms of chlorine element using scanning electron microscope (Zeiss EV018) equipped with an EDS. The analysis was conducted in mapping mode at 20.0 kV. The samples were sputter-coated with gold for enhanced conductivity.

Contact Angle (CA) Measurement – Static contact angles of deionized water or diiodomethane on PDMS-based coating was measured with a Theta Auto 113 (KSV NIMA, Biolin) by depositing a droplet of liquid (3 μL) on the sample surface using the sessile method. Five different regions from one single sample were measured to obtain an average value. Surface free energy was determined by using the Owens-Wendt-Rabel-Kaelble method.^[23]

Dynamic mechanical analysis (DMA) – DMA measurements were performed in tensile mode on a TA Q800 instrument. The sample (15 mm long, 7 mm wide, and 0.5 mm thick) was analyzed at the multistain method with the amplitude from 5 to 50 μm at 25 $^{\circ}\text{C}$. Triplicates of each PDMS-based coating were tested to obtain an average value of storage modulus, which represented the bulk modulus.

Surface elastic modulus measurement – The elastic modulus of the surface of PDMS-based coatings were measured by atomic force microscopy (AFM) using Hertzian contact theory, which correlates the AFM force signal (F) to the indentation depth of the spherical tip (δ) and elastic modulus (E) by the following equation^[24]

$$F = \frac{4}{3} \frac{E}{(1-\nu^2)} \sqrt{R\delta^3}$$

where ν is the Poisson's ratio (assumed to be 0.5 for all coatings), R is radius of the spherical tip, and δ is the indentation depth.

Pseudobarnacle removal test – Removal strength of pseudobarnacles on PDMS-based coating was determined according to ASTM D5618 (2011). Aluminous cylindrical studs (10 mm in diameter, 10 mm in height) were glued to the coating surface using two-part epoxy adhesive (Xinda New Material). The adhesive was hardened for 3 d under ambient conditions before testing. The force gauge (SUNDOO SH-500) was moved in parallel to the surface and the required shear force was recorded.

Leachate toxicity assay – Toxicity of the possibly leached substance including unreacted telomer, catalyst, or residual solvent was analyzed with diatom (*Navicula incerta*) growth test. PDMS-based coating on epoxy panel ($1 \times 1 \text{ cm}^2$) was immersed in 5 mL of Guillard's F/2 culture medium for 14 d with vigorous shaking. Then, 300 μL of diatom suspension with a chlorophyll content of $0.50 \mu\text{g mL}^{-1}$ was added to 3 mL of the F/2 medium with leached substance or fresh F/2 medium as control, and the diatom was cultured at 23 $^{\circ}\text{C}$ with a 12:12 h light-dark cycle for 7 d. Afterward, chlorophyll was extracted from the diatom with mixture of dimethyl sulfoxide and acetone (9/1, v/v), and the chlorophyll content was determined by optical density test.

Mass loss test – Before immersion, the weight of uncoated panel (W_p) and initial weight of PDMS-based coating together with panel (W_0) were recorded. The coating was immersed in ASW at room temperature. After certain immersion time, the panel was removed from ASW and carefully washed with deionized water and the dry weight was recorded (W_t). The mass loss of coating was calculated by $(W_0 - W_t)/(W_0 - W_p) \times 100\%$. For each polymer, triplicate samples were tested in parallel to give an average mass loss.

Bacterial biofilm growth test – Marine bacteria *Pseudomonas* sp. NOV776 cells were streaked on MA agar plates (18.7 g marine broth 2216, and 7.5 g marine agar in 500 mL Millipore-Q water) and grown at 30 $^{\circ}\text{C}$ for 48 h. Then, a monoclonal colony was inoculated into a tube containing 3.0 mL of MB2216 medium and grown at 150 rpm at 30 $^{\circ}\text{C}$ for 24 h. 100 mL of the suspension was into 10 mL of MB2216 medium, further cultured for 12 h to mid-log phase, and diluted to $10^8 \text{ cell mL}^{-1}$ with MB2216. 1 mL of the suspension was added in each well of 24-well microplate with PDMS-based coating on bottom. The bacteria biofilm was grown at 30 $^{\circ}\text{C}$ for 48 h. Afterward, the suspension was taken out and the coating was gently washed to remove the unattached biofilm. The biofilm was fixed with methanol and stained with crystal violet (0.3 wt% in water) for 15 min. The excessive crystal violet was washed with ASW and photos of the biofilm were taken by digital camera. Finally, the crystal violet stain was extracted with glacial acetic acid (33% in water) and optical density was measured at $\lambda = 589 \text{ nm}$.

Digital Holographic Microscopy (DHM) – DHM is based on an in-line configuration integrated with an inverted microscope (model IX-83, Olympus, Japan). A collimated illumination was generated by a light-emitting diode (LED, $\lambda = 505 \text{ nm}$, model M505L3, Thorlabs, US) together with an aperture (diameter $\approx 0.5 \text{ mm}$) and a lens. The focal plane of a $40\times$ microscope objective (NA = 0.6, LUCPLFLN40X, Olympus, Japan) was positioned at 10 μm beneath the bottom surface of the observation chamber. Real-time holograms formed by the

interference between the waves scattered by bacteria and the unscattered incident light were captured. The holograms of 1024×1024 pixels were recorded using a sCMOS camera (model Zyla-5.5-CL3, 6.5 μm per pixel, Andor Technology, UK) at 20 frames per s and streamed into a computer for further data processing. 2400 holograms were captured for 2 min after sealing the observation chamber, and the recording process was repeated several times. DHM experiments were performed at 25 $^{\circ}\text{C}$. The method for data processing was detailed in previous literature.^[25] For each trajectory of bacteria with at least 30 consecutive points, the first 10% of its mean-square displacement (MSD) curve was fitted by $\text{MSD}(\Delta t) = D(\Delta t)^\nu$ to acquire the power index ν (D is the diffusion coefficient). The bacteria which moves with $\nu \geq 1$ is sorted as active bacteria.

Diatom adhesion test – Diatom *Navicula incerta* was kindly offered by State Key Laboratory of Marine Environmental Science (Xiamen University). The diatom was cultivated in Guillard's F/2 culture medium at 23 $^{\circ}\text{C}$ with a 12:12 h light-dark cycle. 1 mL of diatom suspension with a chlorophyll content of $0.50 \mu\text{g mL}^{-1}$ was added to a 24-well microplate, which contains an epoxy panel ($1 \times 1 \text{ cm}^2$) coated with PDMS-based coating (blank epoxy panel used as control) and cultured for 24 h. Then, the coating was taken out and gently washed with ASW to remove the unattached diatom. The attached diatom was observed with a microscope (Scope A1, Zeiss) and counts were made by five random fields of views of each coating. Each sample was tested in triplicates.

Marine field test – PDMS-based coatings on glass fiber reinforced epoxy resin panels ($100 \times 100 \text{ mm}^2$) were immersed in Xiamen Bay, China ($24^{\circ}33'\text{N}$, $118^{\circ}09'\text{E}$) at a depth of 1 m. After a certain period of immersion, the panel was taken out of the sea, gently rinsed and photographed. Then, the panel was washed with water jet (pressure: 2 MPa) and photographed again. The panels were immersed in seawater immediately after the inspection. An epoxy resin panel without any coating was used as control.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

fluorinated polymers, marine anti-biofouling, non-leaching antifoulant, self-stratification, silicone

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